SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I R SARVESH / 192521143 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R SARVESH

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4. A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is –0.6.

For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

7. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration. (5 Marks)

8. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical. (5 Marks)

9. A Genetic Algorithm is used to maximize the function f(x) = 2x + 3 for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence. (5 Marks)

10. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance. (5 Marks)

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not. (5 Marks)

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly. (5 Marks)

14. A Genetic Algorithm is applied to maximize the function f(x) = 3x + 2 for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima.
(5 Marks)

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Perceptron Model – Student Pass or Fail (Q1)

Given:

- Study hours (x1) = 2
- Attendance (x2) = 1
- Weights: w1 = 0.6, w2 = 0.4
- Bias (b) = -0.5

Step 1: Calculate the net input

Formula: Net input = (w1*x1) + (w2*x2) + b

Net input = (0.6*2) + (0.4*1) + (-0.5)

Net input = 1.2 + 0.4 - 0.5 = 1.1

Step 2: Apply the step activation function

- If net input >= 0, output = 1.
- If net input < 0, output = 0.

Since 1.1 >= 0, output = 1.

Conclusion: The student passes, because the perceptron output is 1.

2. Delta Learning Rule – Weight Update (Q2)

Given:

- Initial weight (w) = 0.5
- Learning rate (η) = 0.1
- Input (x) = 2
- Actual output = 0.6
- Target output = 1

Step 1: Calculate the error

Formula: Error = Target output - Actual output

Error = 1 - 0.6 = 0.4

Step 2: Calculate the weight update

Formula: Weight update = η*Error*Input

Weight update = 0.10.42 = 0.08

Step 3: Calculate the new weight

Formula: New weight = Initial weight + Weight update

New weight = 0.5 + 0.08 = 0.58

Conclusion: The updated weight after one iteration is 0.58.

3. Confusion Matrix – Disease Prediction (Q3)

Given:

- True Positive (TP) = 180
- True Negative (TN) = 150
- False Positive (FP) = 20
- False Negative (FN) = 50

Total = TP + TN + FP + FN = 180 + 150 + 20 + 50 = 400

Step 1: Calculate accuracy

Formula: Accuracy = ((TP + TN)/Total)*100

Accuracy = ((180 + 150)/400)*100 = 82.5%

Step 2: Calculate precision

Formula: Precision = (TP/(TP + FP))*100

Precision = (180/(180 + 20))*100 = 90%

Step 3: Calculate sensitivity (recall)

Formula: Recall = (TP/(TP + FN))*100

Recall = (180/(180 + 50))*100 = 78.26%

Step 4: Calculate specificity

Formula: Specificity = (TN/(TN + FP))*100

Specificity = (150/(150 + 20))*100 = 88.24%

Conclusion: The model has 82.5% accuracy and 90% precision. However, its recall is only 78.26%, meaning it misses some patients with the disease. It may be useful as a supporting tool, but further testing and improvement are needed before using it for medical diagnosis.

4. Overfitting in a Deep Neural Network (Q5)

Given:

- Training accuracy increases from 60% to 95%.
- Validation accuracy initially increases from 58% to 72%, then decreases to 68%.

Step 1: Identify the problem

The problem is overfitting.

Step 2: Explain overfitting

Overfitting occurs when a model learns the training data too closely, including noise and unnecessary details. As a result, it performs well on training data but poorly on unseen data.

Step 3: Suggest two techniques

1. Dropout: Randomly disables some neurons during training to prevent the network from depending too heavily on particular neurons.
2. Early stopping: Stops training when validation performance starts getting worse, preventing the model from learning unnecessary details.

Step 4: Importance of validation accuracy

Validation accuracy measures how well the model performs on data that was not used for training. It helps identify overfitting and determine when to stop training.

Conclusion: The model is overfitting because training accuracy increases while validation accuracy decreases. Dropout and early stopping can improve its generalization performance.

5. Perceptron Model – Loan Approval (Q6)

Given:

- Income (x1) = 1.5
- Credit score (x2) = 2
- Weights: w1 = 0.7, w2 = 0.5
- Bias (b) = -0.6

Step 1: Calculate the net input

Formula: Net input = (w1*x1) + (w2*x2) + b

Net input = (0.7*1.5) + (0.5*2) + (-0.6)

Net input = 1.05 + 1 - 0.6 = 1.45

Step 2: Apply the step activation function

- If net input >= 0, output = 1 (Approved).
- If net input < 0, output = 0 (Rejected).

Since 1.45 >= 0, output = 1.

Conclusion: The perceptron predicts that the loan is approved, because the output is 1.